



Syllabus Application



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CS 306 Database Systems

Faculty Faculty of Engineering and Natural Sciences

Semester Spring 2025-2026

Course CS 306 - Database Systems

Time/Place	<u>Time</u>	<u>Week Day</u>	<u>Place</u>	<u>Date</u>
	10:40-11:30	Mon	SBS-G071	Feb 16-May 22, 2026
	11:40-13:30	Wed	FENS-G077	Feb 16-May 22, 2026

Level of course Undergraduate

Course Credits SU Credit:3, ECTS:6, Engineering:6

Prerequisites CS 204 or DSA 201

Corequisites CS 306R

Course Type Lecture

Program Requirements >

Instructor(s) Information

Yücel Saygın

Email: ysaygin@sabanciuniv.edu

Course Information

Catalog Course Description

This course covers the fundamental topics in database management systems including ANSI/SPARC architecture, data abstraction, data independence; data definition and manipulation languages; entity-relationship and relational data models, integrity constraints, query languages like SQL, relational algebra and calculus, functional dependencies, normal forms, transaction processing recovery and concurrency control, data structure for database management systems such as indices, B-trees, hashing.

Course Learning Outcomes:

1.	After the course, the student will be able to develop a database application starting with the conceptual design leading to logical and physical designs.
2.	After the course, the student will be competent on relational model and SQL.
3.	After the course, the student will be able to implement a database application based on a given design.
4.	After the course, the student will be able explain the concept of functional dependency and will be able to avoid redundancies and other problems causing by bad designs.
5.	After the course, the student will be able to explain how a DBMS concurrently handles the transactions and what happens after a system crash.
6.	After the course, the student will be able to explain how the data is physically stored, managed, and queried within the database.

Course Objective

To supply the students with the theory and practice of database management systems, and database application development.

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Course Materials

Resources:

All materials are on sucourse

Technology Requirements:

Laptops

Assessment

The final grade for this course will be based on:

	Percentage (%)	Number of assessment methods
Final	35%	
Midterm	40%	1
Participation	10%	
Group Project	15%	1

Policies

Course Policies:

Check sucourse

AI Policies

[Guidelines on the "Use of Generative AI" for students](#)

SU Academic Integrity Statement:

<https://www.sabanciuniv.edu/en/academic-integrity-statement>

Health Report Requirements

[Student Medical Reports Instruction Letter](#) 

Weekly Content

Week	Topic
Week 1	Basics of Database Tools and Accessing Databases from the Internet
Week 2	Entity Relationship (ER) Model

Week	Topic
Week 3	Relational Model and ER to Relational Model Conversion
Week 4	Relational Algebra
Week 5	From Relational Algebra to SQL
Week 6	Advanced SQL Queries
Week 7	Functional Dependencies
Week 8	Database Normalization
Week 9	Concurrency Control
Week 10	Transaction Management
Week 11	Basics of Data Storage
Week 12	Indexing
Week 13	Database Recovery
Week 14	Query Optimization